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Cognitive and adaptive functioning outcomes after cancer diagnosis in early childhood: A systematic review

Jessica Beamish, Josephine Drijver, Marita Partanen *

Princess Máxima Center for Pediatric Oncology, Heidelberglaan 25, Utrecht 3584 CS, The Netherlands

A R T I C L E I N F O

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A B S T R A C T

Objectives: About one quarter of children who are diagnosed with cancer are under four years old, which is a uniquely critical period in neurodevelopment. Younger age at diagnosis is a known risk factor for poorer long-term cognitive and functional outcomes, indicating that the combined impact of cancer and treatment may disproportionally affect this age group. To date, synthesis of the literature in this area has been limited by heterogeneous findings and methodologies. To help guide clinical practice and future research, this review aimed to describe the cognitive and adaptive functioning outcomes of young children with cancer, as well as summarise the impact of risk factors.

Methods: We conducted a systematic review, according to PRISMA guidelines, of cognitive and adaptive functioning outcomes in cancer patients who were younger than four years at diagnosis. We searched PubMed, EMBASE and PsycINFO and included relevant articles. This review was registered in PROSPERO [CRD42024502629].

Results: There were 5747 records retrieved, and 54 papers were included after full-text review, representing 8083 participants. Long-term impairments in the domains of processing speed, attention, executive and adaptive functioning were reported in brain tumour, haematological and solid tumour groups. Brain tumour survivors also show deficits in IQ. Cranial radiotherapy was the most consistently associated risk factor with poorer cognitive and adaptive functioning. Risk of bias analysis showed most studies were classified as 'strong' quality, but there was variability in study sample size and design.

Conclusion: Children diagnosed with cancer in early childhood are at high risk of cognitive and adaptive functioning deficits, potentially reflecting diffuse neurodevelopmental injury. However, outcomes are highly variable, with limited understanding of risk factors beyond medical treatment. The results support the recommended guidelines for early neuropsychological monitoring for all diagnostic groups to identify vulnerable patients and to implement timely interventions.

1. Introduction

About one quarter of children with cancer are diagnosed before the age of four and owing to modern treatment protocols, about 80 % of these children survive [1,2]. These statistics prompt questions pertaining to the long-term impact of cancer and its treatment on young children. This is especially true because early childhood is recognised as the most crucial period for neurodevelopment, during which the dynamic interaction of genetic and environmental factors lays the biological foundations for lifelong functioning [3–6]. The development of the brain progresses hierarchically, meaning that risk and protective factors in early childhood shape its eventual structure and function later in life [4,7]. Until approximately four to five years old, the brain undergoes rapid dendritic and synaptic growth, marking a phase of increased plasticity and maximal sensitivity to environmental input [7]. Experiences occurring during this period have a disproportionally greater impact on the developing brain, eloquently summarised as a time of unique vulnerability and opportunity [3]. Accordingly, adversity occurring during this period, for example, due to acquired brain injury, poverty or limited social resources, leads to more detrimental cognitive and functional impacts than similar experiences in older children [3,5,7,8].

Younger age at diagnosis is a well-established risk factor for poorer long-term cognitive and functional outcomes in paediatric cancer survivors [9–13]. This indicates that the young child's brain is particularly

* Corresponding author.
E-mail address: m.h.partanen@prinsesmaximacentrum.nl (M. Partanen).

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vulnerable to the negative impacts of cancer, its treatment and consequent missed developmental opportunities. Despite this, there are comparatively few studies that have investigated these outcomes specifically within an entirely young child population. Age at diagnosis is often included as a covariate in studies with wider age ranges, meaning the clinical implications are not specific to the younger developmental period. Studies that have investigated only young children have yielded mixed results. Some suggest there may be different outcome profiles when compared to older children. For example, young children demonstrate cognitive and adaptive functioning impairments that do not improve over time and are relatively independent of traditional risk factors, such as diagnosis and treatment [14–19]. Other studies suggest that, similar to older children, diagnosis and treatment factors are crucial for understanding cognitive and adaptive functioning outcomes. For example, younger children with brain tumours treated with chemotherapy perform comparably to their peers [20–22], whereas those treated with both radiotherapy and chemotherapy perform significantly worse [23,24]. These inconsistent findings, along with the heterogeneity in research methodology, make it difficult to draw cohesive conclusions from the literature.

Prospective studies examining the cognitive impact of cancer in young children often only include a measure of intelligence quotient (IQ) or estimated IQ. This is a coarse average measure of cognitive development and often not valid in this population due to the variability in the component subscales [16,21]. It also results in the systematic underestimation of cognitive sequelae in survivors as not all domains are impacted equally [25]. For example, information processing skills (processing speed, attention, working memory) are the most vulnerable due to the treatment-related white matter changes, compared to more crystalised domains such as language [9–12]. Retrospective studies may look more specifically at individual cognitive domains but typically use clinically referred samples and often combine results from various measures within a domain [16,21,26]. Furthermore, a more comprehensive analysis of multiple cognitive and adaptive domains is limited. Studies also suffer from small sample sizes and variability in treatment approaches, testing time points and inclusion of medical and psychosocial risk factors. This variability limits synthesis of the literature and therefore the opportunity to identify disease, treatment and environmental factors most predictive of cognitive and functional decline.

A structured overview and evaluation of the literature would clarify outcomes, as well as facilitate in guiding treatment, identifying high-risk patient profiles, and opportunities for intervention. We sought to address this by conducting a systematic review to answer the research question: what are the neuropsychological outcomes of children diagnosed with cancer in early childhood? Specifically, we aimed to: 1) describe the cognitive and adaptive functioning outcome profiles of young children diagnosed with and treated for cancer and, 2) summarise the impact of any risk factors on these outcomes.

2. Methods

This systematic review was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Mata-Analysis (PRIMSA) guidelines [27] and the protocol was pre-registered in the international prospective register of systematic reviews (PROSPERO: CRD42024502629).

2.1. Search

Relevant studies were identified through a literature search in PubMed, Embase via Elsevier and PsycINFO via Ovid. Databases were searched from 1st January 1990 until 7th February 2024 (see S1 for full search strings). Language was limited to results published in English.

The search sought to include all available literature on cognitive and adaptive functioning outcomes of children who were diagnosed with cancer before the age of four years. To do this we operationalised our research question with the Population, Intervention/Exposure, Comparison, Outcome (PICO) framework and translated this to appropriate search teams for each database. The following components were included in the search strings: MeSH or other subject terms, synonyms and search filters. All authors and a librarian designed and peer reviewed the search, according to the Peer Review of Electronic Search Strategies (PRESS) guidelines [28]. The goal of implementing the PRESS guidelines is to check accurate translation and adaptation of the research question as well as correct choice and usage of search terms, Boolean operators and filters for the databases searched. The final search included the following key terms, including variations and synonyms: 1) cancer (e.g. "neoplasm"), 2) young child (e.g. "infant") and 3) neuropsychological outcomes (e.g. "cognition"). One author (JB) deduplicated the results and performed forwards and backwards citation analysis. Studies were eligible if they met the following criteria:

1. Original, peer-reviewed, research article.
2. Population: children less than four years old at time of cancer diagnosis. For studies that included patients with a wider age range, the mean age at diagnosis needed to be under four years. For studies that did not note age at diagnosis, the study was included if age at diagnosis could be inferred reasonably as under four years from age at first assessment.
3. Reported on cognitive or adaptive functioning outcomes, at any time after diagnosis, using validated measures [29] and compared performance to normative data or a control group.
4. Sample size of at least 20 patients with cognitive and adaptive outcomes. For, studies that had a sample size > 20 at first assessment time point, but < 20 at later time points, or < 20 in longitudinal analyses, analyses with < 20 were excluded.

2.2. Study Selection

Title and abstract screening was conducted independently by two authors (JB, JD), using ASReview [30]. ASReview is a program which increases efficiency and transparency in the systematic review process by using a machine learning framework to continuously refine articles shown based on prior reviewer decisions. We used the default settings (model: naïve Bayes, query strategy: maximum, feature extraction: term frequency – inverse document frequency) and chose a stopping rule shown to maximise sensitivity and specificity, after a total of 20 % of the records were screened with 5 % consecutively classified as irrelevant [31]. First author (JB) reached this point after screening 52.99 % of records and second author (JD) reached this point after screening 41.62 %. Conflicts were resolved by consensus. After title and abstract screening, the records were imported into Covidence (https://www.covidence.org/), a web-based collaboration software platform that streamlines the production of systematic and other literature reviews, for full text review and data extraction. Full texts were evaluated against the inclusion criteria and discrepancies were resolved by consensus. The screening process is summarised in Fig. 1.

2.3. Data Extraction

Study characteristics and outcome data from each study were recorded in a purpose-built data extraction form in Covidence, which was initially piloted on three studies. Two authors (JB, JD) systematically extracted data from the included studies, and extracted variables such as year of publication, number of patients, age at diagnosis, diagnosis type, measures and time point of assessment. Details regarding the cognitive and adaptive functioning outcomes were extracted either qualitatively or quantitatively, depending on the data available within the study.

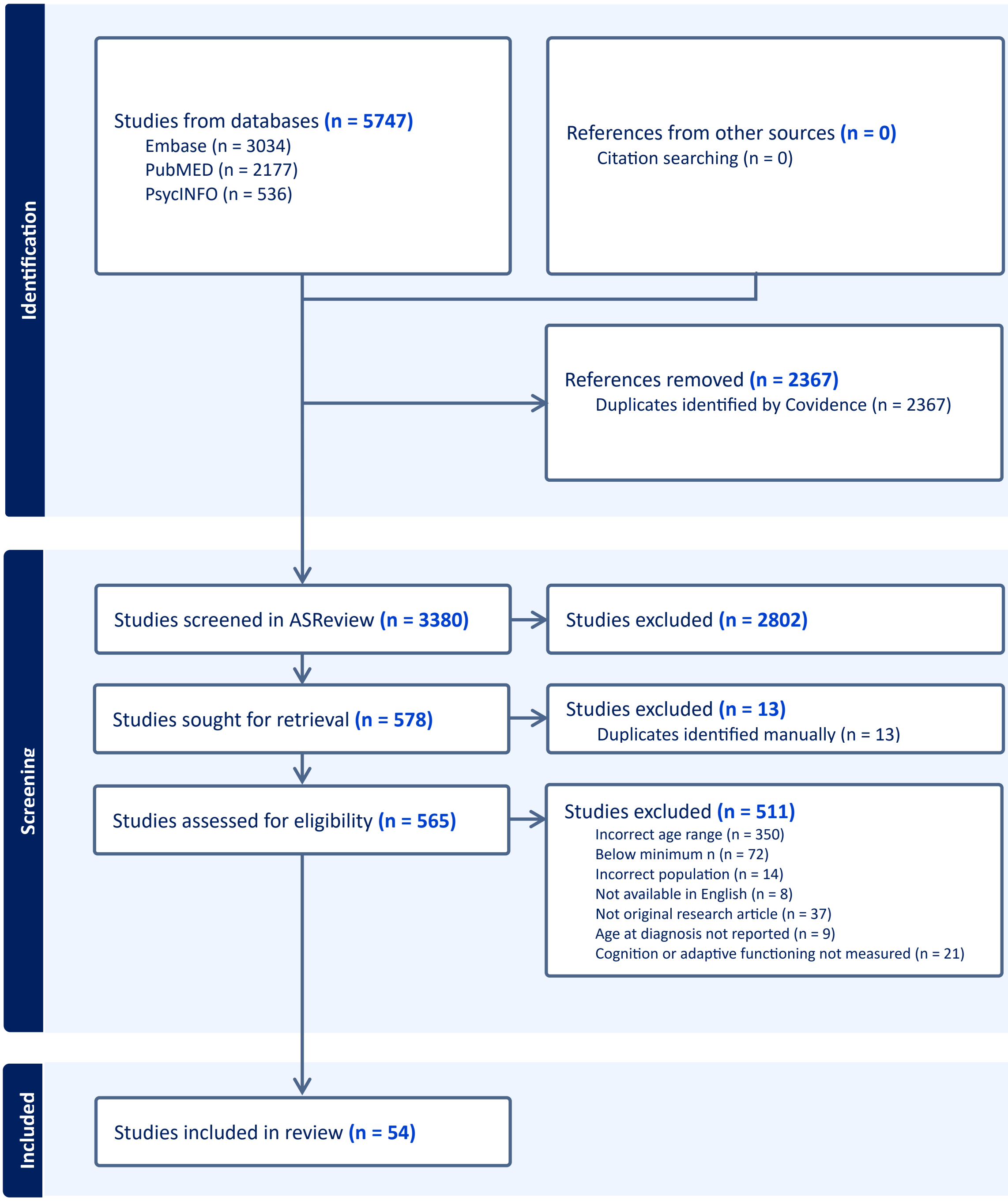

Fig. 1. PRISMA diagram of the systematic review screening process.

2.4. *Quality Assessment*

Two authors (JB, JD) independently assessed the risk of bias for a sub-section of 10 studies, compared the results and resolved conflicts by consensus. First author (JB) then rated the risk of bias for remaining studies. We used the QualSyst tool for critical appraisal of studies in health science [32]. The QualSyst tool provides a systematic, reproducible and quantitative means of assessing the methodological quality of research over a broad range of study designs. Total score is based on eight criteria: appropriate study design and research question, definition of outcomes and exposures, reporting of bias and confounding factors, and sufficient reporting of results and limitations. Criteria can be answered as ‘yes’ (2), ‘partial’ (1), ‘no’ (0), and ‘NA’. A QualSyst score > 80 % is interpreted as strong quality, 60–79 % as good quality, 50–59 % as adequate quality, and < 50 % as poor methodological quality.

3. Results

3.1. *Study Characteristics*

The database search of PubMed, Embase and PsycINFO identified 5747 articles. After de-duplication, title and abstract screening, full text screening and forward and backward citation analysis, 54 articles fulfilled the inclusion criteria. See Fig. 1 for the PRISMA diagram. Characteristics and relevant results of each included article are presented in Tables 1 – 3 and S4 - 5.

Most studies used a prospective cross-sectional design (50 %;

Table 1
Summary of cross-sectional outcomes for brain tumour diagnoses.

First Author	Year	Country	Study design	N	Dx	Age at dx (years)	Age at ax (years)	Treatment	Outcome Domains	DQ/IQ	Other Cognitive Domains	Adaptive Functioning	Significant Risk Factors	Non-significant Risk Factors	Study Quality Assessment
ACUTE															
Ali [14]	2021	USA	Prospective longitudinal	108	Mixed brain tumours	M = 1.7 SD = 1.0	Not noted	Surgery; Chemotherapy; Focal radiotherapy; Whole brain or craniospinal radiation	Developmental functioning IQ Attention Working Memory Executive functioning Adaptive functioning	↓ ↓	n.s. n.s. n.s.	↓	Lower SES, younger age at diagnosis	Other clinical and demographic factors	0.91
Merchant [50]	2014	USA	Prospective longitudinal	76	Infratentorial ependymoma	Med = 3.3	Not noted	Surgery; Chemotherapy; Focal radiotherapy	Developmental functioning IQ Memory	↓ ↓	n.s.		VP shunt placement, younger age at radiation	Pre-irradiation chemotherapy	0.82
Sands [51]	2010	USA	Prospective longitudinal	51	Malignant brain tumours	M = 2.99 SD = 2.08	M = 3.52	Surgery; Chemotherapy; Focal radiotherapy; Whole brain or craniospinal radiation	Developmental functioning IQ Visuomotor Language Memory	n.s. n.s.	n.s. n.s. n.s.				0.82
Stargatt [52]	2006	Australia	Prospective longitudinal	40	Mixed brain tumours	Not noted	M = 2.67, SD = 1.37,	Surgery; Chemotherapy; Focal radiotherapy; Whole brain or craniospinal radiation	Adaptive functioning			↓		Sex, tumour location, age at diagnosis	0.95
LONG-TERM															
Ali [14]	2021	USA	Prospective longitudinal	65	Mixed brain tumours	M = 1.7 SD = 1.0	Not noted, 2 years post dx	Surgery; Chemotherapy; Focal radiotherapy; Whole brain or craniospinal radiation	Developmental functioning IQ Attention Working Memory Executive functioning Adaptive functioning	↓ ↓	↓ ↓ ↓	↓			0.91
Ashley [60]	2012	Australia, USA, Canada	Prospective longitudinal	74	Medulloblastoma	Not noted	Med = 4.8	Surgery; Chemotherapy; Focal radiotherapy; Whole brain or craniospinal radiation	Developmental functioning	↓					0.82
Fouladi [24]	2005	USA	Retrospective longitudinal	67	Mixed brain tumours	M = 1.79	> 5 after diagnosis	Surgery; Chemotherapy; Focal radiotherapy;	Developmental functioning IQ	↓ ↓				Chronic seizures, severe ototoxicity	0.82

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Table 1 (*continued*)

First Author	Year	Country	Study design	N	Dx	Age at dx (years)	Age at ax (years)	Treatment	Outcome Domains	DQ/ IQ	Other Cognitive Domains	Adaptive Functioning	Significant Risk Factors	Non-significant Risk Factors	Study Quality Assessment
								Whole brain or craniospinal radiation							
Morrall [54]	2019	UK	Prospective cross-sectional	59	Ependymoma	M = 2 SD = 0.8	Primary group: M = 14.79 SD = 3.95, Relapsed group: M = 15.62 SD = 3.34	Surgery; Chemotherapy; Focal radiotherapy; Whole brain or craniospinal radiation	IQ Processing speed Working memory Memory Adaptive functioning	↓	↓ n.s. n.s.	↓	Relapsed disease		0.91
Sands [51]	2010	USA	Prospective longitudinal	26	Malignant brain tumours	M = 2.99 SD = 2.08	M = 6.38 SD = 1.93	Surgery; Chemotherapy; Focal radiotherapy; Whole brain or craniospinal radiation	Developmental functioning IQ Visuomotor Language Memory	n.s. n.s.	n.s. n.s. n.s.		Greater time since diagnosis	Age at diagnosis, intravenous MTX	0.82
Stargatt [52]	2006	Australia	Prospective longitudinal	21	Mixed brain tumours	Not noted	M = 5.54 SD = 0.41	Surgery; Chemotherapy; Focal radiotherapy; Whole brain or craniospinal radiation	IQ Visuomotor Memory Adaptive functioning	↓	↓ ↓	↓	Younger age at diagnosis, greater time since diagnosis	Tumour location, treatment, age at radiotherapy, time since radiotherapy	0.95
Copeland [20]	1999	USA	Prospective longitudinal	21	Posterior fossa tumour	M = 1.83	M = 9	Surgery; Chemotherapy; Whole brain or craniospinal radiation	Developmental functioning IQ Visuomotor Language Memory Executive functioning	n.s. n.s.	↓ n.s. ↓ n.s.		Radiotherapy	Time since diagnosis, tumour recurrence, extent of resection, site of tumour, survival	0.77
Dhall [53]	2020	USA/ Canada	Prospective cross-sectional	24	Medulloblastoma	M = 2.35 SD = 1.48	M = 7.28 R	Surgery; Chemotherapy; Focal radiotherapy	IQ Processing speed Working memory Memory Adaptive functioning	n.s.	↓ n.s. n.s.	↓		Cumulative MTX, radiotherapy	0.77
Fay-Mc Clymont [21]	2017	USA and Canada	Retrospective cross-sectional and longitudinal	24	Medulloblastoma	M = 2.45 SD = 1.13	M = 6 SD = 1.8	Surgery; Chemotherapy; Focal radiotherapy; Whole brain or craniospinal radiation	IQ Processing speed Visuomotor Attention Working memory Visuospatial Language Memory	n.s.	↓ ↓ ↓ ↓ ↓ ↓ ↓		Hearing problems, radiotherapy, posterior fossa tumours	Sex, age at diagnosis, histology, metastatic status, extent of resection, high dose MTX, intrathecal chemotherapy, age at assessment	0.73

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Table 1 (*continued*)

First Author	Year	Country	Study design	N	Dx	Age at dx (years)	Age at ax (years)	Treatment	Outcome Domains	DQ/IQ	Other Cognitive Domains	Adaptive Functioning	Significant Risk Factors	Non-significant Risk Factors	Study Quality Assessment
									Executive Functioning Adaptive Functioning		↓	↓			
Heitzer [15]	2019	USA	Retrospective cross-sectional	25	Low grade glioma	Med = 0.52	M = 9.9	Surgery; Chemotherapy; Focal radiotherapy; Whole brain or craniospinal radiation	IQ Visuomotor Attention Memory Executive functioning Adaptive functioning	↓	↓ (Visuomotor) ↓ (Attention) n.s. (Memory) ↓ (Executive functioning)	↓	Number of chemotherapy regimens, more tumour directed surgeries, visual field constriction	NF1 status, CSF shunt placement, sensorineural hearing loss, sex, radiation, time from diagnosis	0.77
Jurbergs [16]	2019	USA	Retrospective cross-sectional	79	Mixed brain tumours	M = 2.39 SD = 1.46	M = 4.52 SD = 1.20	Surgery; Chemotherapy; Focal radiotherapy; Whole brain or craniospinal radiation	Developmental functioning IQ Attention Adaptive functioning	↓ (Developmental functioning) ↓ (IQ)	↓ (Attention)	↓	Younger age at diagnosis, greater time since diagnosis	Sex, treatment status (on vs. off), radiotherapy, radiotherapy type (focal vs. craniospinal, photon vs. proton), hearing loss, history of posterior fossa syndrome	0.73
Lacaze [57]	2003	France	Prospective cross-sectional	27	Optic pathway glioma	Med = 1.3	Med = 8.7;	Chemotherapy; Focal radiotherapy	IQ Working memory Visuospatial Memory Executive functioning	n.s.	n.s. (Working memory) n.s. (Visuospatial) ↓ (Memory) ↓ (Executive functioning)		Radiotherapy, NF1 diagnosis, greater tumour volume	SES, age at diagnosis, extent of disease, visual acuity	0.73
Liu [58]	2019	USA	Retrospective cross-sectional	21	Low grade glioma	M = 6.47	M = 10.5 R	Surgery; Chemotherapy; Focal radiotherapy; Whole brain or craniospinal radiation; No treatment	IQ Adaptive functioning	↓		↓	Number of chemotherapy regimes		0.73
Moore [22]	1992	USA	Prospective cross-sectional	28	Mixed brain tumours	M = 1.58;	M = 8	Surgery; Chemotherapy; Focal radiotherapy; Whole brain or craniospinal radiation	IQ Visuomotor Working memory Language Memory	n.s.	↓ (Visuomotor) ↓ (Working memory) n.s. (Language) ↓ (Memory)		Cranial radiation, younger age at diagnosis, greater time since radiotherapy		0.68
O'Neil [55]	2020	USA and Canada	Prospective cross-sectional	43	Mixed brain tumours	M = 2.85, SD = 1.97	M = 7.97	Surgery; Chemotherapy; Focal radiotherapy; Whole brain or craniospinal radiation	IQ Processing speed Working memory Memory Adaptive functioning	↓	↓ (Processing speed) n.s. (Working memory) n.s. (Memory)	↓		Treatment regimen (Head Start III regime D vs. D2), brain tumour diagnosis, radiotherapy, sex, ethnicity, age at diagnosis, time since diagnosis	0.73

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Table 1 (*continued*)

First Author	Year	Country	Study design	N	Dx	Age at dx (years)	Age at ax (years)	Treatment	Outcome Domains	DQ/ IQ	Other Cognitive Domains	Adaptive Functioning	Significant Risk Factors	Non-significant Risk Factors	Study Quality Assessment
Ottensmeier [56]	2020	Germany	Prospective cross-sectional	72	Medulloblastoma and ependymoma	Med = 2.3	Med = 7.5	Surgery; Chemotherapy; Focal radiotherapy; Whole brain or craniospinal radiation	IQ Visuomotor Attention Executive functioning Adaptive functioning	n.s.	 ↓ n.s. ↓	 n.s.	Craniospinal radiation	Age at surgery, sex, time from start of treatment	0.73
Wagner [59]	2020	UK	Prospective cross-sectional	118	Posterior fossa tumour	M = 3.38 SD = 1.11	M = 34.8, SD = 10.1	Surgery; Chemotherapy; Focal radiotherapy; Whole brain or craniospinal radiation	IQ	↓			Radiotherapy	Brain tumour diagnosis	0.77
Lafay-Cousin [116]	2016	Canada and USA	Retrospective cross-sectional	24	Medulloblastoma	M = 2	Med = 6	Surgery; Chemotherapy; Focal radiotherapy; Whole brain or craniospinal radiation	IQ Processing speed Working memory	n.s.	 n.s. n.s.				0.55

Note. Dx, diagnosis; Ax, assessment; M, mean; Med, median; SD, standard deviation; R, range; N, number of participants in analyses; DQ, developmental quotient; IQ, intellectual quotient; ↓, cancer patients performed significantly below normative data or controls; n.s., the difference between cancer patients and normative data or controls was not significant; VP, ventriculoperitoneal; CSF; cerebrospinal fluid; SES; socioeconomic status; MTX, methotrexate; NF1, neurofibromatosis type 1. Tests used by each study and domain classification can be found in S2-3.

n = 27), others used prospective longitudinal (22 %; n = 12), retrospective cross-sectional (22 %; n = 12) and retrospective longitudinal (4 %; n = 2) designs. One study (2 %) used a cross-sectional design with a combination of retrospective and prospective data collection [33]. Articles included predominantly North American patient groups (76 %; n = 41), with others from Europe (16 %; n = 9), Australia (4 %; n = 2), Asia (2 %; n = 1) and the Caribbean (2 %; n = 1). The mean age at diagnosis of patients was 2.53 years, with a minimum of 0.52 years [15] and a maximum of 3.90 years [34–36]. There was a wide range in mean age at assessment from 22.23 months [37] to 33 years [38]. The total number of patients in each study ranged from 22 [39] to 3085 [40]. The most frequently examined cancer diagnostic group was brain tumours (35 %; n = 19), followed by haematological malignancies (31 %; n = 17) and solid tumours (15 %; n = 8). Two studies (4 %) examined a combination of haematological and solid diagnoses, and eight studies (15 %) examined all diagnoses. Common exclusion criteria included pre-existing diagnoses of a neurodevelopmental or psychiatric disorder (41 %; n = 22) and limited fluency in the primary language of the country where the study was conducted (28 %; n = 15). The total number of patients represented in this review was 8083. Cognitive and adaptive functioning outcomes were reported at various time points from diagnosis onward.

3.2. *Outcome measures*

A wide variety of cognitive and adaptive outcome measures were utilised, totalling 66 different tests, in the included studies. For this review, the tests were categorised into the following outcome domains: developmental quotient (DQ)/IQ, other cognitive domains (which included tests of processing speed, visuomotor skills, attention, working memory, visuospatial, language, memory and executive functioning), and adaptive functioning. The full list of measures as well as the domain categorisation is available in S2–3. The most frequently employed measures were the Bayley Scales of Infant Development [41], age-appropriate Wechsler tasks [42–45], Beery-Buktenica Developmental Test of Visual-Motor Integration [46], the Behaviour Rating Inventory of Executive Function [47], the Vineland Adaptive Behaviour Scales [48] and the Adaptive Behaviour Assessment System [49]. Overall, 45 (83 %) studies included a measure of DQ/IQ, 33 (61 %) studies included a measure of at least one other cognitive domain and 24 (44 %) studies included a measure of adaptive functioning. The outcome measures used by each included study are available in S3.

3.3. *Study Quality Assessment*

The quality of most included studies was classified as ‘strong’ (50 %; n = 27) or ‘good’ (44 %; n = 24). Three studies (6 %) were classified as ‘adequate’, and none were classified as ‘poor’. The average quality assessment rating for studies included was 80.08 %. For studies that had lower ratings, common sources of potential bias included: the methods of subject selection, small sample size and missing effect size estimates for the main outcomes/results. Quality ratings for each study are indicated in Tables 1 – 4 and S3–4.

3.4. *Cognitive and adaptive functioning outcomes*

For the purposes of this review and to enhance potential clinical utility of the findings, the results are summarised by cancer diagnosis category. Brain tumours consists of studies which examined outcomes of children with brain tumours, haematological malignancies consists of those with blood cancer and solid tumours consists of those with extracranial solid tumours. There were also several studies which included patients from different diagnostic categories (e.g., reported combined outcomes of brain tumour and haematological patients), these studies are included in the mixed diagnoses category. We have also dichotomously classified the timing of outcome assessment as the “acute” or

Table 2
Summary of cross-sectional outcomes for haematological diagnoses.

First Author	Year	Country	Study design	N	Dx	Age at dx (years)	Age at ax (years)	Treatment	Outcome Domains	DQ/ IQ	Other cognitive domains	Adaptive Functioning	Significant Risk Factors	Non-significant Risk Factors	Study Quality Assessment
ACUTE															
Tremolada [61]	2019	Italy	Prospective cross-sectional	44	ALL	Not reported	M = 4.52 SD = 1.06	Chemotherapy	Adaptive functioning			↓	Higher number of treatment toxicities	Maternal education, days of hospitalisation	0.55
LONG-TERM															
Edelmann [65]	2014	USA	Prospective cross-sectional	39	ALL	M = 2.81 SD = 1.73	M = 26.71 SD = 3.44	Chemotherapy; Whole brain or craniospinal radiation	IQ Visuomotor Attention Working Memory Memory Executive functioning	n.s.	 ↓ ↓ ↓ ↓ ↓		Radiotherapy		0.86
Kim [67]	2015	Korea	Prospective cross-sectional	42	ALL	M = 3.8 SD = 2.3	M = 10.5 SD = 2.4	Chemotherapy; Whole brain or craniospinal radiation	IQ Attention Executive Functioning	n.s.	 ↓ ↓		Radiotherapy		0.88
Kunin-Batson [34]	2014	USA	Prospective cross-sectional	263	B ALL	M = 3.9	M = 13.1	Chemotherapy	IQ Processing speed Visuomotor Attention	n.s.	 n.s. n.s. n.s.				0.91
Seghatol-Eslami [72]	2024	USA	Retrospective cross-sectional	50	ALL	M = 3.36 SD = 2.20	M = 11.06 SD = 3.18	Chemotherapy	Adaptive functioning			↓	Male sex, area deprivation index, worse parent rated executive functioning	Executive functioning, neighbourhood-specific variables, academic support	0.91
Spiegler [33]	2006	Canada	Prospective and retrospective cross-sectional	79	High-risk ALL	M = 2.83 SD = 1.07	M = 13.3 SD = 2.7	Chemotherapy; Whole brain or craniospinal radiation	IQ Attention Language Memory	↓	 ↓ ↓ ↓		Radiotherapy		1.00
van der Plas [62]	2020	Canada	Prospective cross-sectional	71	ALL	M = 3.8 SD = 1.7	M = 11.9 SD = 2.8	Chemotherapy	IQ	↓				Treatment protocol, age at diagnosis, sex	0.95
Viola [36]	2017	USA	Prospective cross-sectional	256	ALL	M = 3.9 SD = 1.8	M = 12.8 SD = 2.5	Chemotherapy	Executive functioning		↓		ADHD diagnosis		1.00
Waber* [68]	2012	USA, Canada and Puerto Rico	Prospective cross-sectional	128	ALL	Med = 3.58	M = 9.17	Chemotherapy; Whole brain or craniospinal radiation	IQ Attention Visuospatial Memory Executive functioning	n.s.	 ↓ ↓ n.s. n.s.		High-risk disease		0.91
Waber* [71]	2007	USA, Canada and Puerto Rico	Prospective cross-sectional	79	Standard-risk ALL	Med = 3	Med = 9	Chemotherapy; Whole brain or craniospinal radiation	IQ Processing speed Visuospatial Memory	n.s.	 ↓ n.s. n.s.			Type of treatment	0.81

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Table 2 (*continued*)

First Author	Year	Country	Study design	N	Dx	Age at dx (years)	Age at ax (years)	Treatment	Outcome Domains	DQ/ IQ	Other cognitive domains	Adaptive Functioning	Significant Risk Factors	Non-significant Risk Factors	Study Quality Assessment
									Executive functioning Adaptive functioning		n.s.	 ↓			
Hardy [35]	2017	USA	Prospective cross-sectional	89	High-risk B ALL	M = 3.9 SD = 2.1	M = 5.1	Chemotherapy	IQ Processing speed Working memory	n.s.	 n.s. n.s.		Younger age at diagnosis, US public insurance		0.79
Jain [66]	2009	USA	Retrospective cross-sectional	103	ALL	M = 3.9	M = 11.4	Chemotherapy	Working memory Attention Executive functioning		↓ ↓ ↓		High-risk disease	Cumulative intravenous MTX, age at diagnosis	0.77
Kadan-Lottick [69]	2009	USA	Retrospective cross-sectional	91	Standard-risk B ALL	M = 3.2 – 3.4 SD = 1.0 – 1.4	M = 13.0 – 13.2	Chemotherapy	IQ Processing speed Visuomotor Attention Working memory Memory	n.s.	 n.s. n.s. n.s. n.s. n.s.		Female sex, younger age at diagnosis		0.75
Kaleita [73]	1999	USA	Prospective cross-sectional	30	ALL	M = 0.63 SD = 0.26	M = 5.18 SD = 1.43	Chemotherapy	Developmental functioning	n.s				Parent occupation, parent education, sex	0.77
Krull [63]	2014	USA	Prospective cross-sectional	38	ALL	M = 2.8 – 2.9 SD = 1.59 – 1.63	M = 25.4 – 27.5 SD = 2.57 – 3.52	Chemotherapy; Whole brain or craniospinal radiation	IQ Processing speed Attention Memory Executive Functioning	↓	 ↓ ↓ ↓ ↓			Radiation dose	0.77
Reddick [64]	2006	USA	Prospective cross-sectional	28	ALL	M = 3.1 SD = 2.3	M = 11.1 SD = 2.6	Surgery; Chemotherapy; Whole brain or craniospinal radiation	IQ Attention	↓	 ↓		Smaller white matter volume		0.77
Waber [70]	2000	USA	Prospective cross-sectional	23	ALL	M = 3.7 SD = 1.8	M = 7.8 SD = 1.6	Chemotherapy; Whole brain or craniospinal radiation	IQ Visuospatial	n.s.	 ↓		Dexamethasone treatment		0.73

Note. Dx, diagnosis; Ax, assessment; M, mean; Med, median; SD, standard deviation; R, range; N, number of participants in analyses; DQ, developmental quotient; IQ, intellectual quotient *, used the same sample; ↓, cancer patients performed significantly below normative data or controls; n.s., the difference between cancer patients and normative data or controls was not significant; VP, ventriculoperitoneal; SES; socioeconomic status; MTX, methotrexate; NF1, neurofibromatosis type 1. Tests used by each study and domain classification can be found in S2-3.

Table 3
Summary of cross-sectional outcomes for solid tumour diagnoses.

First Author	Year	Country	Study design	N	Dx	Age at dx (years)	Age at ax (years)	Treatment	Outcome Domains	DQ/ IQ	Other cognitive domains	Adaptive Functioning	Significant Risk Factors	Non-significant Risk Factors	Study Quality Assessment
ACUTE															
Willard [76]	2014	USA	Prospective longitudinal	84	Retinoblastoma	M = 1.29 SD = 1.12	Not noted	Surgery; Chemotherapy	Developmental functioning Adaptive functioning	n.s.		 n.s.			0.91
LONG-TERM															
Hesko [79]	2023	USA	Retrospective cross-sectional	837	Neuroblastoma	M = 1.95 SD = 2.64	M = 25.03 SD = 4.87	Chemotherapy; Focal radiotherapy; Whole brain or craniospinal radiation	Processing speed Attention Memory Executive Functioning		↓ ↓ ↓ ↓		Platinum exposure, cranial radiation, hearing loss, neurological conditions, chronic conditions		0.91
Notteghem [75]	2003	France	Prospective cross-sectional	76	Mixed extracranial tumours	Med = 3.6	Med = 15.7	Surgery; Chemotherapy; Focal radiotherapy; HSTC	IQ Visuospatial Memory	n.s.	 n.s. n.s.		Maternal education level, auditory deficit, younger age at BMT	Diagnosis, conditioning regimen, number of BMTs, hormonal disturbances, school absence, paternal education	0.82
Tonning Olsson [38]	2019	USA	Prospective cross-sectional	158	Wilms tumour	M = 3.6 SD = 2.6	M = 33 SD = 9.1	Surgery; Chemotherapy; Focal radiotherapy	IQ Processing speed Attention Working memory Memory Executive functioning	↓	 n.s. n.s. ↓ ↓ ↓		Race, musculoskeletal conditions, neurologic conditions		0.91
Willard* [78]	2017	USA	Prospective longitudinal	84	Retinoblastoma	M = 1.30 SD = 1.13	M = 5	Surgery; Chemotherapy	Developmental functioning Adaptive functioning	↓		 n.s.	Higher parent stress		0.86
Willard* [77]	2021	USA	Prospective longitudinal	73	Retinoblastoma	M = 1.33 SD = 1.27	M = 10.86 SD = 1.08	Surgery; Chemotherapy	Developmental functioning IQ Adaptive functioning	n.s. n.s.		 n.s.	Enucleation only treatment		0.82
Willard* [76]	2014	USA	Prospective longitudinal	84	Retinoblastoma	M = 1.29, SD = 1.12	M = 5	Surgery; Chemotherapy	Developmental functioning Adaptive functioning	↓		 n.s.			0.91
Ek [39]	2002	Sweden	Prospective longitudinal	21	Retinoblastoma	M = 0.75	M = 6	Surgery; Focal radiotherapy	Developmental functioning IQ	n.s. n.s.			Unilateral disease		0.64
Foster [74]	2022	USA	Prospective cross-sectional	246	Wilms tumour	Med = 3.0	Med = 30.5	Surgery; Chemotherapy; Focal radiotherapy	Processing speed Attention Memory Executive functioning		↓ n.s. ↓ ↓		Race	Treatment exposures	0.77

Note. Dx, diagnosis; Ax, assessment; M, mean; Med, median; SD, standard deviation; R, range; N, number of participants in analyses; DQ, developmental quotient; IQ, intellectual quotient *, used the same sample; ↓, cancer patients performed significantly below normative data or controls; n.s., the difference between cancer patients and normative data or controls was not significant; VP, ventriculoperitoneal; SES; socioeconomic status; MTX, methotrexate; NF1, neurofibromatosis type 1. Tests used by each study and domain classification can be found in S2-3.

Table 4
Summary of cross-sectional outcomes for mixed diagnoses.

First Author	Year	Country	Study design	Dx	N	Age at dx (years)	Age at ax (years)	Treatment	Outcome Domains	DQ/IQ	Other cognitive domains	Adaptive Functioning	Significant Risk Factors	Non-significant Risk Factors	Study Quality Assessment
ACUTE															
Bornstein [37]	2012	Italy	Prospective cross-sectional	Mixed non-CNS cancers	122	M = 1.58 SD = 0.93	M = 1.85 SD = 0.99	Surgery; Chemotherapy	Developmental functioning Visuomotor Language	↓	 ↓ ↓		Lower adaptive functioning	Child behavioural status, medical variables, child language	0.91
Quigg [80]	2013	USA	Prospective longitudinal	Mixed cancers	30	M = 2.24 SD = 1.06	At diagnosis	Surgery; Chemotherapy	Developmental functioning	↓					0.90
Kenney [17]	2020	USA	Retrospective longitudinal	Mixed cancers	29	M = 0.99 SD = 0.61	M = 1.97 SD = 0.55	Surgery; Chemotherapy; Focal radiotherapy; Whole brain or craniospinal radiation; HSTC	Developmental functioning Adaptive functioning	↓		 ↓	Brain tumour diagnosis	Treatment status, sex	0.77
Zucchetti [19]	2022	Italy	Prospective cross-sectional	Mixed non-CNS cancers	53	M = 1.83 SD = 0.83	At diagnosis	Chemotherapy	Developmental functioning	↓					0.73
LONG-TERM															
Park [82]	2018	Canada	Prospective cross-sectional	Mixed cancers	25	M = 2.60 SD = 1.43	M = 29.29 SD = 8.98	Surgery; Chemotherapy; Focal radiotherapy; Whole brain or craniospinal radiation; Immunotherapy; HSTC	Memory		↓				0.95
Shin [40]	2023	USA	Retrospective cross-sectional	Mixed cancers	3085	M = 3.8	M = 31.9 SD = 8.3	Surgery; Chemotherapy; Focal radiotherapy; Whole brain or craniospinal radiation	Processing speed Attention Memory Executive Functioning (domains combined to represent overall functioning)	n.s.			High physical, somatic and psychologic symptom burden, cranial radiation		0.91
Willard [18]	2017	USA	Retrospective cross-sectional	Mixed cancers	98	M = 2.52 SD = 1.52	M = 5.17 SD = 0.54	Surgery; Chemotherapy; Focal radiotherapy; Whole brain or craniospinal radiation; HSTC	Developmental functioning IQ Processing speed Working memory Adaptive functioning	↓ ↓	 ↓ ↓	 ↓	Male sex	Treatment type, cancer diagnosis	0.86
Harman [26]	2019	USA	Retrospective cross-sectional	Mixed cancers	61	M = 2.62 SD = 1.32	M = 5.00 SD = 0.72	Surgery; Chemotherapy; Focal radiotherapy; Whole brain or craniospinal radiation; Immunotherapy; HSTC	IQ Executive functioning Adaptive functioning	↓	 ↓	 ↓		Cancer diagnosis, treatment status	0.77
Howell [81]	2017	USA	Prospective cross-sectional	Mixed cancers	78	M = 2.6	M = 12.7 SD = 1.1	Surgery; Chemotherapy; Focal radiotherapy; Whole brain or craniospinal radiation	IQ Executive functioning	n.s.	 n.s				0.75

(*continued on next page*)

Table 4 (*continued*)

First Author	Year	Country	Study design	Dx	N	Age at dx (years)	Age at ax (years)	Treatment	Outcome Domains	DQ/IQ	Other cognitive domains	Adaptive Functioning	Significant Risk Factors	Non-significant Risk Factors	Study Quality Assessment
Coombs [83]	2014	Trinidad and Tobago	Prospective cross-sectional	Mixed cancers	26	M = 2.54	M = 6	Surgery; Chemotherapy; Focal radiotherapy	Developmental functioning	↓			ALL diagnosis, male sex	Treatment type, duration of treatment, age at treatment initiation/completion, time since treatment	0.55

Note. Dx, diagnosis; Ax, assessment; M, mean; Med, median; SD, standard deviation; N, number of participants in analyses; CNS, central nervous system; DQ, developmental quotient; IQ, intellectual quotient; ↓, cancer patients performed significantly below normative data or controls; n.s., the difference between cancer patients and normative data or controls was not significant; VP, ventriculoperitoneal; SES; socioeconomic status; MTX, methotrexate; NF1, neurofibromatosis type 1; ALL, acute lymphoblastic leukaemia. Tests used by each study and domain classification can be found in S2-3.

"long-term" period. The acute outcome period was defined as from diagnosis up until one year after treatment completion. The long-term outcome period was defined as from one year after treatment completion onwards.

3.5. *Brain tumours*

Nineteen studies reported on cognitive and adaptive functioning outcomes of young children with brain tumours. Eighteen studies reported long-term outcomes and four studies reported outcomes in the acute period [14,50–52].

3.5.1. *Cross-sectional outcomes*

In terms of acute outcomes, three studies measured IQ. Two of these found that patients with cancer performed significantly poorer compared to normative or control data [14,50], while the other did not find a significant difference [51]. No differences were reported in memory, visuomotor, attention, working memory, language and executive functioning [14,50,51]. Adaptive functioning was measured in two studies, both of which found significantly poorer performance in children with cancer [14,52].

There were 18 studies that reported long-term IQ outcomes. The majority (n = 10) found significantly reduced IQ in brain tumour survivors when compared to controls. Cognitive assessments covering multiple domains were conducted in 14 studies. The most consistently reduced domains in cancer survivors compared to controls were processing speed (n = 4/5) [21,53–55], visuomotor performance (n = 6/7) [15,20,22,52,56] and executive functioning (n = 5/6) [14,15,21,56,57]. Four studies measured language and most (n = 3/4) [20,22,51] found no difference between cancer survivor and control performance. Mixed results were reported in the domains of memory, attention and visuospatial skills, see Table 1 for details. Adaptive functioning was measured in 10 studies, nine of which found significant reductions in cancer survivors [14–16,21,52–55,58].

A variety of risk factors were examined in relation to long-term cognitive and adaptive functioning, with demographic or medical factors being the most consistently studied. Age at diagnosis was examined in eight studies, with three finding younger age at diagnosis was associated with poorer outcomes [16,22,52], five did not find significant associations [21,51,55–57]. Eight studies investigated time since diagnosis or treatment, with four studies indicating poorer performance associated with longer time [16,22,51,52], and four studies not finding a significant association [15,20,55,56]. Cranial radiation exposure was examined in nine studies with the majority (n = 6) finding associations with poorer outcomes [20–22,56,57,59], whereas three did not find a significant relationship [15,53,55]. Two studies examined treatment with proton vs. photon radiation, both reported no difference in outcomes [14,16]. Treatment with methotrexate was examined in three studies and found no increased risk with high-dose, intrathecal or cumulative dose [21,51,53]. Other risk factors were examined less frequently, see Table 1.

3.5.2. *Longitudinal outcomes*

Five studies examined the trajectory of IQ over time in brain tumour survivors and met sample size inclusion criteria. One of these found significant decline in IQ over time [24], while four found stable performance [14,50,51,60]. Longitudinal results for other cognitive domains were infrequently reported: one study identified declining executive and attention functioning over time, but stable working memory [14] and one reported declining language functioning [50]. Longitudinal adaptive outcomes were in one study which reported stable functioning [14].

The most consistently examined risk factor in longitudinal studies in brain tumours was cranial radiation treatment (n = 4), where two studies found that radiation was associated with declines in IQ over time [24,50]. The other two studies did not find this association [14,60].

Ventriculoperitoneal (VP) shunt placement was examined in three studies, with one reporting a significant association between VP shunt and decline over time [14] and two reporting no association [24,50]. Two studies reported on tumour location, both found supratentorial location associated with decline over time [14,24]. Treatment with chemotherapy was not associated with decline in cognitive and adaptive functioning over time in three studies [14,51,60]. Other risk factors were examined infrequently, a summary is available in S4.

3.6. Haematological malignancies

Seventeen studies examined outcomes in children diagnosed with haematological malignancies, all of which were in acute lymphoblastic leukaemia (ALL) populations. One reported results in the acute period, which was at the end of treatment [61]. The authors found overall decreased adaptive functioning, with increased risk for children who had a greater number of treatment-related toxicities, for example neuropathy or nausea.

Most studies of haematological malignancies ($n = 16$) reported on long-term outcomes and included a measure of IQ ($n = 12$). Eight indicated that IQ was not significantly different from normative or control data. In contrast, there were four studies that indicated significantly poorer performance in survivors [33,62–64]. Studies employing comprehensive cognitive assessments were less common ($n = 8/16$). In general, results suggested that survivors consistently performed below normative or control groups on domains of attention, visuomotor processing and executive functioning [33,36,63–68]. There was inconsistent evidence for processing speed, working memory, visuospatial skills and memory [34,35,63,65,66,68–71]. Language was assessed in one study, which found poorer performance in survivors [33]. Long-term adaptive functioning was assessed by two studies, both of which indicated decreased adaptive functioning in ALL survivors compared to normative or control data [68,72]. See Table 2 for a summary.

Cranial radiotherapy treatment and high-risk disease were consistent risk factors for poorer cognitive and adaptive outcomes in ALL, see Table 2. Other risk factors were less consistent. Younger age at diagnosis was examined in four studies, with mixed results. Some studies suggested young age was [35,69] or was not [62,66] a significant risk factor. The impact of patient sex was also inconsistent. Two studies did not find a significant association [62,73], one found male sex [72], and one found female sex [69], was associated with higher risk of cognitive problems. Two U.S. studies examined socioenvironmental risk factors and found that children from areas of higher deprivation and those with public health insurance had poorer cognitive and adaptive functioning [35,72].

3.7. Solid tumours

Eight studies examined outcomes of children with solid tumours, four of which used longitudinal designs. All studies reported long-term outcomes, but one of the longitudinal studies also reported outcomes in the acute phase. Three of these performed comprehensive cognitive assessments [38,74,75].

3.7.1. Cross-sectional outcomes

Acute outcomes were reported by one study [76], which reported no differences between retinoblastoma patients and controls on the domains of general cognitive and adaptive functioning.

Long-term cognitive outcomes were generally mixed. Of the six studies which measured IQ, three found that solid tumour survivors' performance was not different from norms [39,75,77], while the other found survivors performed significantly worse [38,76,78]. In terms of the other cognitive domains, executive functioning was consistently reduced in survivors across studies [38,74,79]. Processing speed, attention and memory were also generally poorer in survivors, but with more mixed findings [38,74,79]. Working memory was measured in one study which found significantly poorer performance in cancer survivors [38]. Visuospatial skills were reported in one study and findings were not significant [75]. Visuomotor and language were not measured in solid tumour survivors. Adaptive functioning was measured in three studies of the same cohort, each of which reported no significant difference between cancer survivors and their peers [76–78]. A summary of results is presented in Table 3.

Race was identified as a significant risk factor for poorer cognitive outcomes in two studies in solid tumours [38,74], as were chronic health conditions [38,79] and hearing loss [75,79]. Other risk factors that were identified by one study each were: lower maternal education [75], cranial radiation and platinum exposure [79], see Table 3.

3.7.2. Longitudinal outcomes

Four studies examined cognitive and adaptive functioning over time, all of these examined retinoblastoma patients, three of which were from the same patient cohort [76–78], see S5. The results of the series on the same cohort indicated IQ and adaptive functioning declined over time until the age of five. Then, from five until 10 years old, the trajectory changed, and general cognitive and adaptive functioning increased over time. The other study reported stable IQ over time in cancer survivors [39].

Risk factors associated with decline over time in these studies were: treatment with enucleation, non-white race and higher parent stress. Non-significant factors included sex, socioeconomic status and age at diagnosis.

3.8. Mixed tumour diagnoses

Ten studies reported outcomes of children with mixed diagnoses, see Table 4. Two of these examined the combined outcomes of haematological and solid groups and eight examined combined outcomes of all diagnostic groups. Four studies reported outcomes within the acute timeframe, all of which examined IQ. These studies reported that cancer patients performed significantly lower than controls [17,19,37,80]. Bornstein et al., [37] also examined visuomotor and language and found significant reductions for cancer patients. Adaptive functioning was measured by Kenney et al., [17], who reported significantly poorer functioning in cancer patients.

In terms of long-term outcomes, five studies included a measure of IQ, which indicated cancer survivors performed significantly lower than controls. Two studies measured executive functioning and reported divergent results. Harman et al., [26] found significantly poorer executive functioning in cancer survivors, whereas Howell et al., [81] did not find a significant difference between groups. Processing speed, working memory and memory domains were measured by one study each and both reported poorer functioning in cancer survivors [18,82]. Adaptive functioning outcomes were reported in two studies, and both found this to be reduced in cancer survivors [18,26].

Type of cancer diagnosis was examined as a risk factor in four studies with mixed results. For acute outcomes, brain tumour diagnosis was identified as risk factor for poorer cognitive, but not adaptive outcomes [17]. In the long-term, ALL diagnosis was identified as a significant risk factor in one study [83] and two studies did not identify differences between diagnostic groups [18,26]. Male sex was identified by two studies as a significant risk factor for poorer long-term [18,83], but not acute, outcomes [17]. Two studies found that lower adaptive functioning, cranial radiation, and larger somatic, physical and psychological symptom burden were associated with poorer cognitive outcomes [37,40].

Two studies examined the impact of being on active cancer therapy and found no difference between those on- and off-treatment [17,26]. Similarly, therapy type (radiotherapy vs. chemotherapy) was not associated with outcomes in two studies [18,83]. Acute medical variables, child behaviour, therapy duration, age at treatment initiation and time since treatment completion were not associated with outcomes [37,83].

See Fig. 2 for a summary of cross-sectional results.

4. Discussion

This study aimed to systematically review the literature on cognitive and adaptive functioning outcomes of children diagnosed with cancer before four years of age. We identified and synthesised the results of 54 original, relevant, peer-reviewed articles. Overall, the results indicate that this population, regardless of diagnostic category, is at high risk for cognitive and adaptive functioning deficits. Long-term impairments in the domains of processing speed, attention, executive and adaptive functioning were identified in all diagnostic groups. Brain tumour survivors also exhibit short- and long-term deficits in IQ. Cranial radiotherapy was the most consistently associated risk factor with poorer cognitive and adaptive functioning across diagnostic groups. There were limited studies addressing outcomes in the acute period but indicated cognitive and adaptive deficits may become evident early in the diagnostic and survivorship trajectory. Therefore, the results of this review underscore the importance of ongoing neuropsychological monitoring of young children with all cancer diagnoses.

4.1. Early survivorship

Comparatively few studies (20 %) reported cognitive and adaptive outcomes in the acute period (during, and up to one year after treatment). From the studies that did, children with brain tumours exhibit reduced IQ and adaptive functioning compared to their peers. Children with ALL demonstrated reduced adaptive functioning and those with

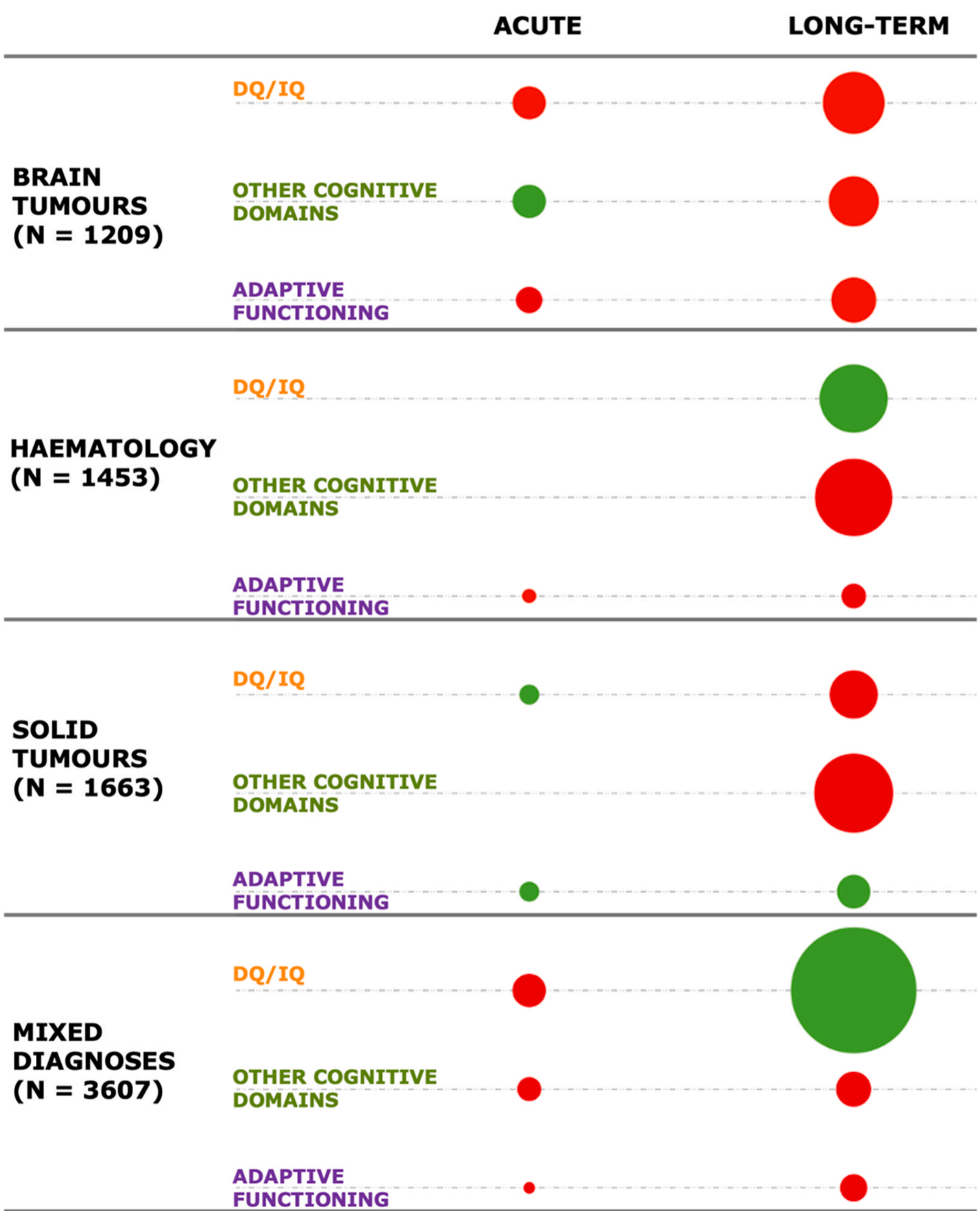

Fig. 2. Summary of cross-sectional cognitive and adaptive functioning results. Note. Acute time frame refers to the time frame from diagnosis up to 1 year after treatment completion, long-term refers to any time longer than 1 year after treatment completion. N represents number of overall participants in each diagnostic group. Circle size represents number of participants in that cell. Circle colour denotes comparison of cancer patients to normative or control data, where the final colour is determined by the category with the highest number of participants. Red, significantly below norms or controls; Green, not significantly different; DQ, developmental quotient; IQ, intellectual quotient. Tests used by each study and domain classification can be found in S2–3.

solid tumours demonstrated intact IQ and adaptive functioning. Studies of mixed diagnostic groups reported reductions in IQ, and other cognitive and adaptive domains. These results indicate that children with cancer are vulnerable to deficits early in the diagnosis and treatment trajectory. The timing of these outcome measurements may be too early to identify treatment-related impacts and may instead reflect the developmental impact of the disease itself. This is consistent with a study in older children, where 20 % of children with brain tumours and 4 % of children diagnosed with non-CNS cancer show cognitive impairments at diagnosis [84]. The authors hypothesised the deficits observed in the brain tumour patients were due to tumour infiltration disrupting the development of neural networks. Further, the deficits observed in the non-CNS cancer patients could be due to impact on the CNS of cancer-induced immune inflammatory responses and their regulation by neuroendocrine pathways [84,85].

4.2. Long-term survivorship

Long-term outcomes were more commonly reported, with the results spanning from one year after treatment completion until mid-adulthood. Interestingly, outcome profiles were relatively similar across diagnostic groups. Overall, young childhood cancer survivors most frequently experience deficits in information processing speed, attention, executive functioning and adaptive functioning, when compared to healthy peers. This is consistent with findings from the wider paediatric oncology literature where certain domains have shown differential vulnerability to disease and treatment, including processing speed, attention, and executive functioning, and are termed “core deficits” [9–12,86]. Therefore, these deficits appear to impact children across the age range. Neuroimaging studies indicate the mechanism underlying these impairments is diffuse brain injury. Grey and white matter structure, connectivity of cortical and subcortical areas, and associated functional activation have been shown to be affected in long-term of survivors of all types of childhood cancer [87]. Chemotherapy has been posited as one of the etiological factors that may underlie this diffuse brain injury. Both animal and human studies have demonstrated neurotoxicity, via direct and indirect mechanisms, which may be detrimental to the developing brain after chemotherapy [88].

Long-term adaptive functioning deficits were also identified in survivors diagnosed in early childhood, a pattern that is consistent with the literature in older children [89,90]. This could be linked to the executive functioning deficits discussed above as these are consistently related to poor adaptive functioning [12,91]. Executive functioning encompasses skills such as initiation, working memory, planning, organising, behaviour and self-monitoring. These are critical life skills for meeting daily occupational and interpersonal expectations across settings. Poor executive functioning can limit day to day performance, even in the context of generally age-appropriate IQ [91,92]. Further, when accounting for psychological symptoms and fatigue, executive dysfunction is the most predictive factor of adaptive deficits in brain tumour survivors [93].

In addition to these “core deficits”, studies of brain tumour survivors consistently reported poorer long-term IQ when compared to their peers. ALL survivors were not different from their peers and evidence for solid tumour survivors was inconsistent. This vulnerability of brain tumour survivors diagnosed at a young age is consistent with literature in older children, as brain tumours and related treatments disrupt neurodevelopment and have a cascading impact on general cognitive outcomes [86,94–96]. Perhaps surprisingly, the trajectory of IQ over time appears to be stable. The results indicated young children diagnosed with brain tumours experience a significant negative early impact on IQ that does not significantly decline over time. This is different from the trajectory described in the wider paediatric brain tumour literature, where IQ does decline over time [96,97]. The stable pattern found in this review was, however, also identified and discussed by Ali et al., [14]. The authors hypothesised that young children experience a significant initial impact from their brain tumour and surgery, but less appreciable impact over time, perhaps due to the plasticity of the young brain. Other explanations for these differences could include the possibility that IQ is not a sensitive measure to the cognitive skills most vulnerable to decline, or that treatment protocols minimise the use of cranial radiotherapy in young patients, which has been consistently linked to cognitive decline [86,96,97].

4.3. Risk factors

Consistent with previous literature, treatment with cranial radiotherapy was the most consistently associated risk factor with decline in both cognitive and adaptive functioning domains, across diagnostic groups [10,86,89,90,95,96,98–102]. As radiation is delivered, a fraction of this energy is inevitably deposited along the radiation beam’s path and immediately adjacent to the target site, causing damage to healthy brain tissue. This causes a diffuse lesion which is characterised by degradation of the subcortical white matter that does not resolve with time [103]. High-risk disease or disease that required more aggressive treatment, was a less consistent risk factor for poor long-term outcomes. Measurement of other risk factors was infrequent, with conflicting results even within diagnostic and treatment groups. Interestingly, some of the factors with inconsistent findings in this age group are well-established risk factors in the wider literature, such as, age at diagnosis, sex and treatment with methotrexate. This suggests that risk profiles for poorer outcomes in young children might be different to that of older children. Most studies examined demographic or medical risk factors, with social and environmental risk factors being included less frequently. There was some evidence for poorer outcomes in children with lower socioeconomic status (SES), higher neighbourhood deprivation, non-white race, and no insurance coverage in the US. This is consistent with growing evidence in this area. For example, studies which investigated the impact of SES on developmental outcomes in brain tumour survivors showed a greater decline in cognition over time for children from low SES and greater economic hardship backgrounds [98,104,105].

4.4. Strengths and limitations

This systematic review has several strengths, including providing a comprehensive and exhaustive overview of the literature using transparent, rigorous, and reproducible methods. By way of evidence classification, the results of this review provide at least level 2 evidence regarding the cognitive and adaptive functioning outcomes of child cancer survivors diagnosed at a young age [106]. This is the first systematic review in this area and clearly highlights the cognitive and functional vulnerability of young children with all types of cancer. The risk of bias and methods assessment, conducted according to PRISMA guidelines, demonstrates the varying quality and heterogeneity of the literature. We used ASReview [30], an AI tool, to streamline initial the process title and abstract screening. These types of tools may be helpful for future researchers, especially as evidence and the need for systematic reviews in the field grows.

There are also limitations to acknowledge when considering the generalisability of the findings. First, there was vast heterogeneity in the outcomes of the included studies, making it difficult to draw conclusions. This likely resulted from the methodical differences across studies, such as in: inclusion criteria, measures, assessment timing and risk factors measured. Further, the paediatric oncology field consistently has small sample sizes, which can lead to varying results. While we tried to address this with our sample size inclusion criterion, this secondarily led to several relevant studies being excluded from this review. Second, most of the included studies were conducted in the United States and excluded participants who did not speak English or were diagnosed with an additional cognitive vulnerability (e.g. neurodevelopmental disorder). Therefore, the results of this review may reflect the outcome of this population but may not be generalisable to patients with culturally and linguistically diverse backgrounds, a sub-group that may indeed be more vulnerable. Third, this review included studies published from 1990 onwards. While these results are relevant for current long-term survivors, treatment protocols have changed substantially in this period. Therefore, the results are potentially less representative of those children who have been treated recently or with newer treatments, for example, with proton radiation or immunotherapy.

4.5. Future perspectives

This review highlights multiple areas of need in the literature. There were few studies examining cognitive and adaptive functioning outcomes in the acute period or utilising longitudinal designs. Those that used longitudinal designs tended to focus on children with brain tumours and only included measures of IQ. As a result, the acute cognitive and adaptive impacts of cancer diagnosis is relatively unknown, particularly the trajectory of the most vulnerable cognitive domains (information processing, attention and executive functioning) over time. Addressing these limitations would help in identifying potential decline and provide the basis for interventions to possibly change the trajectory, as early and timely hospital-based interventions have been shown to be effective in infants with cancer [19]. Further, most of the included studies were in brain tumour and ALL populations and therefore outcomes for diagnoses that have a high prevalence in young children (e.g. neuroblastoma) are widely understudied. In terms of methods, 29 % (n = 16) of included studies used retrospective samples, which included children who were clinically referred and therefore may bias results. Future research should focus on employing prospective designs to ensure outcomes and risk factors of interest are followed closely.

There is a clear need for identification of specific risk factors for this age group, as treatment and demographic factors (apart from cranial radiotherapy) seem to have lower predictive power in this population. Future research designs should focus on studying young children as an individual group and widen their focus to include socioenvironmental risk factors. Socioenvironmental factors, such as poverty and family functioning, have been shown to consistently influence neurodevelopment in a wide range of populations [7]. Another risk factor that was not addressed in the included studies is repeated anaesthesia exposure. There is some evidence to suggest that greater anaesthesia exposure is a risk factor for poorer cognitive outcomes in paediatric cancer populations [107–109]. This is particularly relevant for young patients, as their brains may be more vulnerable [110]. Further, this age group often requires anaesthesia for many cancer-related procedures, leading to greater cumulative exposure. Understanding the role of these factors in paediatric cancer can facilitate identification of the most vulnerable patients.

4.6. Implications

This review provides the basis for some clinical recommendations. First, young children from all diagnostic groups are at risk of developing cognitive and adaptive functioning deficits and therefore neuropsychological screening should be implemented earlier in treatment and survivorship. Second, measures of IQ (or estimated IQ) are not sensitive to the cognitive and functional domains most vulnerable to decline in these patients. In line with previously published monitoring guidelines, a battery which includes measures of processing speed, attention, executive and adaptive functioning should be used in clinical care and research trials to gain a representative overview in this population [111–113]. While there are clear challenges with measuring these outcomes in the youngest children with valid and reliable tools, there are global efforts to address this need, such as the NIH Baby Toolbox and the D-score [114,115]. Another option is to use parent-report questionnaires as the same measure can be used irrespective of the child's age. Third, almost all included studies noted variability in individual patient outcomes and coupled with the limited understanding of specific risk factors, this indicates the continued need for an individualised approach to patient care. Implementation of these recommendations will allow children vulnerable to developing cognitive and adaptive deficits to be identified earlier, provide the basis for proactive intervention and potentially mitigate the trajectory of decline.

5. Conclusion

In summary, this review sought to evaluate and synthesise the literature on the cognitive and adaptive functioning outcomes, including the impact of risk factors, of children who were diagnosed with cancer in early childhood. The results indicated that this population is at high risk for deficits. Specifically, all diagnostic groups exhibited long-term impairments in the domains of processing speed, attention, executive and adaptive functioning, perhaps reflecting diffuse neurodevelopmental injury. Children diagnosed with brain tumours also have deficits in IQ. Cranial radiotherapy was the most consistent risk factor associated with poorer cognitive and adaptive functioning across diagnosis groups, whereas other risk factors were inconsistent. The results highlight the importance of early neuropsychological monitoring, including of specific cognitive domains, for all diagnostic groups to identify vulnerable patients and to enable timely intervention.

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CRediT authorship contribution statement

Jessica Beamish: conceptualisation, methodology, data curation, investigation, visualisation, writing- original draft, writing – review & editing. **Josephine Drijver:** methodology, investigation, supervision, writing – review & editing. **Marita Partanen:** conceptualisation, funding acquisition, supervision, writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ejcped.2025.100237.

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